

2024



Conduit Power Slashed SCADA Deployment Time from 18 Months to 6 Months with Hitachi Energy

Learn how independent power producer Conduit Power implemented a resilient cloud-based SCADA system using AWS.

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2–3x lower annual costs
than on premises



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Overview

Independent power producer (IPP) [Conduit Power](#) designs, builds, and operates distributed power plants combined with analytics and intelligence. The company needed a supervisory control and data acquisition (SCADA)

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A small purple speech bubble icon with a white outline, indicating a chat or contact function.

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system that was flexible and resilient, but an on-premises solution would be cost prohibitive and burden the operations team.

So, the company began looking for cloud solutions and decided to build a SCADA system on Amazon Web Services (AWS). Now, Conduit Power has a resilient and scalable SCADA system that's efficient, scalable, and cost efficient on AWS.



Opportunity | Using Amazon EC2 to Run a Cloud-Based Network Manager SCADA System for Conduit Power

Founded in 2022, Conduit Power is an IPP that provides customized full-service solutions to help commercial and industrial customers lower power procurement costs and improve the uptime and reliability of delivered power. With its small team, the company wanted to overcome energy challenges for both large enterprises and small customers. To manage its industrial control systems while maintaining high resilience and reliability and complying with industry regulations, it needed a scalable SCADA solution.

Conduit Power's team didn't want to spend resources building and maintaining an on-premises system. It began looking for a cloud solution and engaged [Hitachi Energy](#), an AWS Partner and provider of advanced real-time solutions. "We can deploy our Network Manager SCADA system on premises or in the cloud," says Oscar Tillberg, head of sales network control at Hitachi Energy. "After doing an evaluation with Conduit Power, we determined that it was best for the customer to deploy it in the cloud."

The companies chose to build and manage the new SCADA system on AWS, which Conduit Power had already used to analyze billions of rows of power and pricing data and make decisions about building distribution networks. "We value reliability of service, and we already knew that AWS is a reliable cloud provider. We value the flexibility and scale of being able to operate our own systems using virtual private networks for our customers."



than having to manage our own data center operations,” says Matt Herpich, managing director and CEO of Conduit Power.

The project kicked off in January 2024. Hitachi Energy worked with [SnapSoft](#), a cloud-based software development company and [AWS Partner](#), to build and deploy the infrastructure for Conduit Power using [Amazon Elastic Compute Cloud](#) (Amazon EC2), secure and resizable compute capacity for virtually any workload. Instead of starting from scratch, it adapted a SCADA solution that it had already designed for large companies and scaled it down for the customer’s needs. And because they didn’t have to procure and configure new hardware and software, the companies were able to implement the new solution in just 6 months instead of the typical timeline of 12–18 months.

Solution | Reducing Costs by More than 50 Percent Using AWS

Because the solution is scalable on AWS, Conduit Power can focus on growth without worrying about spending millions of dollars on infrastructure. And with Hitachi Energy’s solution already designed to be used by much larger companies, it simply adapted it for Conduit Power’s size. The system includes the best technology, most efficient practices, and built-in compliance with North American Electric Reliability Corporation Critical Infrastructure Protection (NERC CIP) requirements while providing room to grow. The solution also maintains a secure connection on the network using [Amazon Virtual Private Cloud](#) (Amazon VPC), which gives businesses full control over their virtual networking environment. Conduit Power has everything it needs in the cloud—efficiency, reliability, and compliance—and it’s scalable for what the customer will need as it grows. “Because we’re using AWS, we’ve opened up a level of flexibility that’s much needed,” says Herpich.

Building a SCADA solution often requires a multimillion-dollar budget. With affordability a priority for Conduit Power, the decision to build on AWS paid off. Hitachi Energy scaled down its Network Manager SCADA solution in the cloud so that it would meet Conduit Power’s needs without overburdening it with high upfront and ongoing costs. Conduit Power deployed the new system while it is spending one-half to one-third of what these deployments would typically cost on premises. In addition, its annual costs are at least 50 percent lower than they would be on premises.

To meet the high availability needs of the power industry, Hitachi Energy includes a multilayer standby system in its Network Manager SCADA solution. “We deploy the solution with two layers of backups to achieve at least 99.99 percent availability,” says Henry Phan, senior account manager at Hitachi Energy. As a result, Conduit Power’s small team can respond rapidly to any challenges. “This solution is adaptable and flexible. It complies with regulatory requirements and runs in the cloud, which makes it efficient for a company of this small size to manage its own assets,” says Herpich.

By deploying a cloud SCADA system, Conduit Power has achieved the flexible system it needs to serve a wide range of customers without having to manage and adapt every aspect of the infrastructure. The system can scale up for large companies or down for smaller IPPs. And it can do this with high availability. I really don’t



know of a more available service than AWS. We're taking advantage of the resiliency that's built into AWS and using best-in-class cloud cybersecurity policies to overlay authentication practices and prevent phishing and other known risks," says Herpich. "Pairing those two aspects together creates the most resilient system."

Compliance | Architecting to Meet Regulatory Requirements

The electric utility industry in North America must meet the NERC CIP regulatory requirements. These requirements apply to bulk energy systems, which mostly fall under transmission and generation. Under current NERC CIP standards, control systems for bulk energy must be on premises. While there are ongoing efforts to modernize NERC CIP standards to address cloud technologies, Conduit Power needs to have the flexibility in its SCADA architecture to meet NERC CIP requirements. As Conduit Power rolls out its SCADA deployment and adds more plants to its fleet, its business will grow, and so will the amount of power it controls. This will make the company a bulk energy system owner/operator. AWS, Hitachi, and Conduit Power have architected Conduit Power's SCADA system such that a hybrid on-premises and cloud deployment can be implemented to meet CIP requirements as and when required.

Outcome | Expanding the Network Manager SCADA System to More Power Plants

After first deploying it at one plant, Conduit Power is rolling out the SCADA solution on AWS to the rest of its plants. It will continue implementing the solution and monitoring its progress along with the company's growth. "Using a SCADA system in the cloud was something that we had to do out of necessity, and it was a little bit novel and innovative," says Herpich. "Now, it's a selling point on trying to transition the energy sector into kind of a more software-oriented mindset."

About Conduit Power

Founded in 2022, Conduit Power is an independent power producer that offers customized full-service solutions to help commercial and industrial customers lower power procurement costs and improve the reliability of delivered power.

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